

HAS NATIONAL MISSILE DEFENSE TEST LAUNCHES PROVEN THAT THE
NATIONAL MISSILE SHIELD CAN DEFEND AGAINST A ROGUE MISSILE
ATTACK

by

Alexander Dean Curtis

This Graduate Research project was prepared under the direction of the candidate's
Research Committee Member, Adjunct Professor, Extended Campus,
And the candidate's Research Committee Chair,
Assistant Professor, Extended Campus, and has been approved by the Project Review
Committee. It was submitted to the Extended Campus in partial fulfillment of the
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PROJECT REVIEW COMMITTEE:

Art Rosado, DPA
Committee member

Bobby Welch, Ph.D.
Committee Chair

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ABSTRACT

Researcher: Alexander Dean Curtis

Title: GRP Title: Has National Missile Defense test launches proven that the National Missile Shield can defend against a rogue missile attack.

Institution: Embry-Riddle Aeronautical University

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This research paper will provide the reader with a comprehensive analysis of numerous information sources on the nationally significant National Missile Defense test launches and whether the National Missile Shield can protect the Continental United States against a rogue missile threat/attack. Areas researched include Boeing archives, Air Force Air University materials, Congressional, United States General Accounting Office, Secretary of Defense and Federation of American Scientists documentation. This study was qualitative in nature and used analysis of primary and secondary data in evaluating the research question(s). The importance of this study is to provide further insight into the use of National Missile Defense technologies and research in combating the possible problem of rogue missile threats.

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CHAPTER I

INTRODUCTION

Background of the Problem

With the involvement of the researcher in the \$350 million Joint Chiefs of Staff directed Force Development Evaluation test program and the on-going National Missile defense booster verification test launch program, this was an obvious researcher topic for the researcher. Currently there is a discord between the popular media perception and the reality of how well the National Missile Defense testing program has been progressing. This has led the researcher into further analysis of the National Missile Defense program; and if the National Missile Defense shield can actually defend the Continental United States from a rogue missile threat. Additionally, upon initial research, it was discovered that the system was more intricate than the researcher had first imagined.

The first "ballistic" weapons probably were rocks that cavemen hurled at each other. These "missiles" were followed by sticks fitted with pointed stone heads to make spears and later by wood and "string" devices that propelled smaller wooden shafts through the air. (Smith 1999)

But it wasn't until catapults evolved that "missiles"-which could be anything from large flaming arrows (used by the Chinese in the 11th century) to large stones to a body (infected if possible)-became really useful in warfare. In the west, the Greeks set the early design standards for mechanical catapults, designs on which the Romans relied for their siege warfare machines. (Smith 1999)

The Chinese, who invented gunpowder, were also the first to employ explosives to power missiles and the first, in the 1300s, to fire a multi-stage missile. By the 17th century western military forces were experimenting with "war" rockets. Yet not until World War II were long range (up to 180 miles) rockets-the Nazi V-1-and ballistic missiles-the A-4, renamed the V-2-developed. Although it was possible to counter the noisy, relatively slow V-1, there was virtually no defense against the ballistic V-2. (Smith, 1999)

The armed forces of World War Two were ill equipped to deal with V-1and V-2 rockets. The only defense against the V-1 "buzz-bombs" were Royal Air Force fighter interceptors (if they could interdict the slow flying rocket) or accurately placed anti-aircraft fire. However, the V-2 rocket proved to be the real predecessor of the ballistic missile. The V-2 rocket skirted the earth's atmosphere before it plunged down onto its target in England with little or no warning. The V-2 rocket destroyed over 23,000 homes and killed over 100,000 people during its brief period of use. (Smith, 1999)

The technology employed by these first rockets were primitive, they were inaccurate and carried a light payload. The Allied response besides those listed above was very similar to the response during Desert Storm in 1991; find the launch sites and attempt to interdict them, and put them out of commission. This process was a game of cat and mouse. The allied warplanes would scramble to the location of the SCUD launch and attempt to destroy the mobile launchers before they had a chance to hide. This strategy worked to a point. This proved that an effective counter measure besides air power was needed to defeat the missile threat. Additionally, the Allies attacked the transportation system, storage bunkers, command and control sites and troop

concentrations during Operation Crossbow, which slowed down the launching of these weapons. (Smith, 1999)

Therein started the genesis of the American ballistic missile and the ballistic missile defense program. The United States military had began conducting research on ballistic missile defense systems shortly after the conclusion of World War II. From those early efforts until today, the United States has spent over \$120 billion on missile defense, but has yet to develop a reliable system, that can defend the Continental United States against a missile attack. (Smith, 1999)

The United States Air Force and the United States Army have been on the leading edge of research and design of past, present, and future National Missile Defense weapon systems. In 1949, the United States Air Force developed its only surface to air missile in conjunction with the Boeing Aircraft Company. What came about was a pilot less interceptor that could destroy both bombers and cruise missiles. The Bomarc was an unmanned interceptor that was launched vertically. The Bomarc would climb to altitudes of 50,000 feet, and then rotated to a horizontal/more conventional type attitude. After the Bomarc closed to within 10 miles of its target, the internal radar would guide it until intercept. In 1961 the Super Bomarc (B-Model) was tested against two US Navy Regulus II missiles, intercepting the two targets at Mach 2, 100,000-foot altitude, and 375 miles downrange. The United States Air Force had 242 Bomarc missiles in active duty. The last Bomarc was deactivated on 1 July 1972 (Krause, 1999).

The Nike-Hercules project was started in 1945 as a direct result of a U.S. Army study on antiaircraft guided missile problems. The Nike-Hercules project was put into production to defend against an exaggerated Soviet nuclear bomber threat. In 1968, the

United States Congress wanted to decommission the Nike-Hercules missiles, but due to consternation from European allies it was kept on active duty until the advent of the SAM-D, or better known as the Patriot missile defense system. The last Nike-Hercules was withdrawn from Europe in 1984. A total of 25,000 Nike-Hercules missiles were produced with 863 fielded by the United States (Krause, 1999).

In 1955, the United States Air Force started work on the Thor project, which was a hybrid between a strategic and tactical system. The Thor was originally known as a tactical ballistic missile, and then later designated as an intermediate ballistic missile with a range of 1,000-2,000 miles. Thor was originally intended to counter the proposed Soviet fractioning bombardment system, a space-based weapon proposed in the late 1950s. The Thor missile program became a back seat priority after the Soviet Union launched Sputnik on 4 October 1957 and proved that ballistic trajectory could be achieved with a small to medium payload. (Krause, 1999)

Because of the shortcomings of the Nike-Hercules missiles destroying ballistic missiles in the atmosphere, the U.S. Army began a study in 1959 for development of a high-speed terminal phase interceptor. Concurrently, the newest iteration of the Nike-Hercules program became known as the Spartan, which would become the last US nuclear-tipped Anti-Ballistic Missile system. While the Nike-Hercules provided low-level defense against ballistic missiles/cruise missiles, the Spartan provided high-level defense against ballistic missiles. The two weapons systems promised to provide overlapping nuclear Anti Ballistic Missile coverage. (Krause, 1999)

In 1967, Secretary of Defense Robert S. McNamara announced that president Lyndon B. Johnson decided to deploy the Sentinel ballistic missile defense system. In

1969, President Richard M. Nixon renamed the Sentinel system to the Safeguard System, and realigned the focus of the ABMs to a mission of protecting US Intercontinental Ballistic Missile (ICBM) launch facilities. Test missiles were flown from Vandenberg Air Force Base in California from 1970 to 1973. However, the signing of the Treaty on the Limitation of Anti Ballistic Missile Systems on 26 May 1972, US Anti Ballistic Missile (ABM) systems development slowed to a trickle. On 1 October 1975, the only US ABM site at Grand Forks AFB, South Dakota, became operational with 70 Sprint and 30 Spartan missiles. In 1983 further development of the Sprint and Spartan weapons systems was supplanted by research into conventional ABMs and the project was terminated (Krause, 1999).

On January 1984, the Presidential National Security Decision Directive 119 formally established the Strategic Defense Initiative (SDI) to explore "the possibilities of developing missile defenses as an alternative means of deterring nuclear war." Judge Abraham Sofaer, the State Department legal advisor, presented his 1987 conclusion that the ABM treaty did not prohibit the testing of space-based missile defense systems, including directed energy weapons. Defensive satellites, anti-satellites, and active terminal defense systems were seriously considered and options were explored (Krause, 1999).

Theater missile defense systems were studied concurrently with national defense systems. The Patriot Advanced Capability successfully destroyed another Patriot missile simulating an SS-23 ballistic missile during a test at the White Sands national test Range in 1987. When Iraq invaded Kuwait in August 1990, theater missile defense emerged as a predominant factor in missile defense funding calculus. The Fiscal 1991 Appropriation

Conference Committee Report called for a centrally managed theater missile defense program to be established by the Secretary of Defense. Additionally, funding was to be provided through 1997 and multi-service requirements and participation was planned (Krause, 1999).

When US Forces entered Saudi Arabia in defense of Kuwait, the very real possibility of US troops being attacked by ballistic missiles armed with biological or chemical agents appeared as a surprise particularly when the Soviet Union was not the aggressor. However, US troops were not completely unprepared to deal with the threat of theater ballistic missiles. With a combination of attack options the US could posture toward defensive or offensive capabilities in defending its troop/logistical concentrations against Iraqi Scud missiles. Scud sites and Transport Erector Launchers were targeted during the opening 24-hours of the air campaign. Additionally, PAC-2 missiles provided a necessary back up when attack options failed to eliminate all threats preemptively. Without strong offensive capabilities and defensive antimissile systems that worked effectively, the strong U.S. presence during the Gulf War would have faced serious setbacks and the potential for larger casualties existed (Krause, 1999).

In March 1991, after the Gulf War demonstrated that an effective National Missile Defense / theater missile defense system was necessary to future military operations, further research and development was necessary in order to deploy such systems by the mid-1990s. Additionally, in 1993, the Ballistic Missile Defense Organization (BMDO) was established and was provided \$12 billion to further theater missile defense for the years FY 95 to FY 99. (Krause, 1999)

The USAF expanded its research to the Airborne Laser (ABL) and other attack options tests, which fit into the BMDO charter of providing a catcher's mitt defense to National Missile Defense / theater missile defense, while providing inter-service cooperation and joint attack options against future ballistic/theater missile threats (Krause, 1999.)

Currently, BMDO is testing the Ground Based Mid-Course Interceptor (GBI) as the high level defense against ballistic missile threats from third world/rogue nation states. Due to the rising threats and the recent terrorist attacks upon this nation, the urgency of fielding a viable National Missile Defense /theater missile defense system is of the utmost importance. However as cited in the 1998 Welch Report, the timeliness of the system is contingent upon the successful implementation of applicable hardware and software.

If all goes according to the restructured plan, by June 2000, BMDO will have demonstrated the "feasibility" of a National Missile Defense system but not the "readiness to deploy" of the system. The demonstration of readiness will not come until 2003 at the earliest, when the integrated GBI, i.e. operational version of the booster and the Exo-Atmospheric Kill Vehicle (EKV) is to be demonstrated.

(Welch Report, 1998, p.22)

The current National Missile Defense system has the primary mission to defend the Continental United States against an attack consisting of several ballistic missiles launched from a rogue nation. It would also have the capability to protect the Continental United States from an accidental launch from nuclear powers such as Russia or China. Since President George W. Bush has sent notice to Russia that the United

States plans to withdraw from the ABM Treaty (the United States has withdrawn from the ABM Treaty after research was started). The stringent technical requirements imposed on the research and development of the National Missile Defense system may soon be lifted, allowing for unfettered in-depth research and possibly deployment the National Missile Defense system at an earlier date than anticipated.

The first phase of the National Missile Defense system comprises 100 GBIs at a site in Alaska, with a new X-band radar on Alaska's Shemya Island, probably next to the Cobra Dane phased array radar. Four other new X-band radars are planned for other areas of the globe, including Grand Forks in North Dakota, Thule in Greenland, Fylingdales in England, and Vardo in Norway. Ballistic Missile Early Warning System radars in Greenland and Britain would be updated as would the PAVE PAWS radars on the US east and west coasts. This first phase is divided into two steps. The first step, now planned, is called the C-1 configuration, and is to be deployed by 2005. It will consist of 20 interceptors, five early warning radars (Ballistic Missile Early Warning System radars in Alaska, Greenland, and Britain; and PAVE PAWS radars in Massachusetts and California), X-band ground-based radar, and a battle management command, communication and control system. Step C-1 Prime will add 80 interceptors and radar upgrades in 2007, to bring the total number of interceptors to 100. (Aldridge, 2000)

A second phase to be undertaken prior to 2010 comprises another powerful radar at Grand Forks Air Force Base, North Dakota. Along with the powerful radar will be another 100 interceptors. The National Missile Defense system consists of ground-based interceptors (GBIs) which will be alerted by a interlocking Ground Based Radars (GBR),

other radars and the Space-Based Infrared System (SBIRS) satellite constellation for early detection and tracking of hostile missiles and warheads. The Battle Management Command, Control and Communications System (BM/C3) will coordinate all of these actions. (Aldridge, 2000)

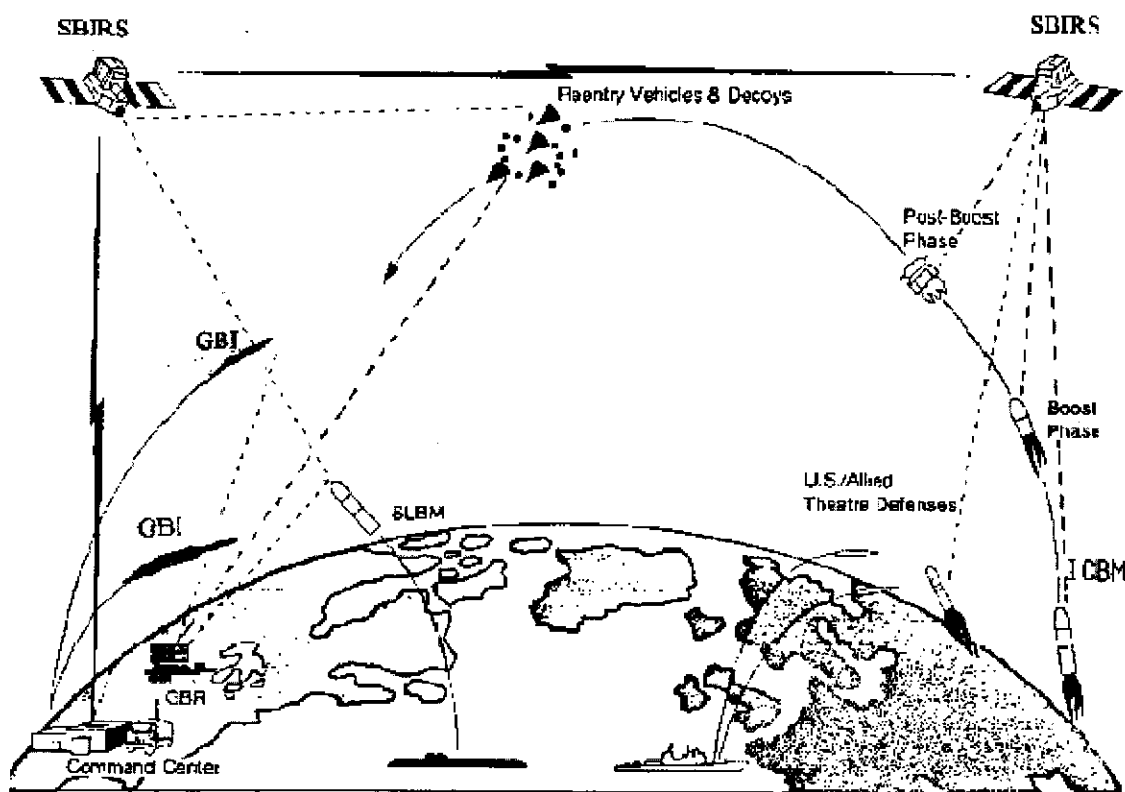


FIGURE 1. Integrated BMDO Scenario. Note. From Pacific Life Research Center (p.28), by Bob Aldridge, 2000, Santa Clara, CA.

The National Missile Defense system has and will be under criticism for the time being due to the findings of the Welch Report in 1998. The Welch Report cited that inadequate testing, spare parts shortage, and management lapses plague the National Missile Defense program. It also criticized the government and contractors for

exhibiting a legacy of over optimism about their ability to develop a reliable missile interceptor (Welch Report, 1998).

However, the Welch Report did not criticize the technology or methodology behind the National Missile Defense program. As with many scholars, it is believed and has been shown through positive testing that the National Missile Defense GBI program will work, and can defend the nation against a rogue missile attack.

The two most recent tests of the GBI during integrated flight tests (IFT) have shown that the EKV, GBR, SBIRS and BM/C3 work well together and have two successful back-to-back kills. National Missile Defense relies upon proven facts that during the trajectory of a ballistic missile during midcourse it is the most vulnerable. Additionally, the National Missile Defense system will use the three-target approach to "killing" missiles. The first target window is during midcourse flight. The second target is during the post-boost phase and lastly, the reentry of the warhead as it enters terminal flight will be targeted. A ballistic missile starts in the boost phase, goes through a midcourse, and ends in terminal flight as it heads towards the target. National Missile Defense tested the midcourse intercept theory during November 2001. This test was conducted successfully on IFT-7. The GBI test program is testing the National Missile Defense system against long-range ICBMs. However there are other systems in place or in research that will deal with the short, medium and IRBMs. The long-range test launches a target vehicle between 5,500 to 10,000 kilometers away from the GBI and tests the connectivity of the integrated system against an ICBM threat. The weapon end of the GBI is the Exo-Atmospheric Kill Vehicle (EKV), which travels at velocities close to 15,000 miles per hour and vectors in on the target vehicle using "hit to kill," which

means that the force of the collision, or kinetic energy that is produced is sufficient enough to destroy the target rather using explosives. This requires very precise computing technology and error free communications between the various National Missile Defense sub-systems (dltranscripts, 2001.)

Researcher's Work Setting and Role

In order to provide the reader with the general academic and professional background of the researcher and its applicability to the topic under research, the following is a highlight of all relevant data:

- Education: Anticipated Master of Aeronautical Science degree, Aviation/Aerospace management, Embry Riddle Aeronautical University, 2002. Bachelor of Art degree, History, California State University at Fresno, 1997.
- Professional Experience: The researcher has spent the last nine years on active duty in the USAF. Four years were spent as a Wideband/Satellite Communication Journeyman. His responsibilities included setting up communications for forward operating locations and communicating with refueling/strike aircraft in a wide range of real world operations and exercises. His responsibilities during the past five years have included working on ICBMs in Montana while holding the following positions: Assistant Officer-in Charge Missile Maintenance Teams Section; Vehicle Control Officer; Officer-in Charge Minuteman Reentry System/Reentry Vehicle Maintenance Section; Commander, Quality Assurance Flight; Commander, Maintenance Support Flight and Commander; Logistics Operations Flight. The researcher is currently completing the USAF Squadron Officers' School via correspondence course.

Statement of the Problem

This research paper will provide the reader with a comprehensive analysis of numerous information sources on the nationally significant National Missile Defense test launches and whether the National Missile Shield can protect the Continental United States against a rogue missile threat/attack. Areas researched include Boeing Archives, Air Force University materials, Congressional documentation, United States General Accounting Office, secretary of Defense and Federation of American Scientists documentation. The study was qualitative in nature and used analysis of primary and secondary data in evaluating the hypothesis of study. The importance of this study is to provide further insight into the use of National Missile Defense technologies and research in combating the possible scenario of defending the United States against rogue missile threats/attacks.

Definition of Terms

Airborne Laser	(ABL)
Anti-Ballistic Missile	(ABM)
Anti-Satellites	(ASAT)
Attack Options	(AO)
Ballistic Missile Defense Organization	(BMDO)
Battle Management Command, Control and Communication Systems	(BM/C3)
Continental United States	(CONUS)
Defensive Satellites	(DSAT)
Exo-Atmospheric Kill Vehicle	(EKV)
Ground Based Mid Course Interceptor	(GBI)

Intercontinental Ballistic Missile	(ICBM)
Intermediate Range Ballistic Missile	(IRBM)
Joint Chiefs of Staff	(JCS)
Limitation of Anti Ballistic Missile Systems	(SALT II)
National Missile Defense	(NMD)
Space Based Infrared System	(SBIRS)
Strategic Defense Initiative	(SDI)
United States Air Force	(USAF)
United States Army	(USA)
Transporter Erector Launcher	(TEL)
Theater Missile Defense	(TMD)

Limitations and Assumptions

There will be some limitations associated with this research, including,

- **Time:** Due to the fact that the researcher is employed on a full time basis as a captain on active duty in the United States Air Force working approximately 55-60 hours per week, the researcher estimates spending 10 hours per week toward the completion of this project for a total of 200 hours.
- **Funding:** Since the researcher is on a limited budget for this project, he is allocating approximately \$200 for telephone, and automobile expenses.
- **Access to information:** It is possible that in pursuit of information gathering, the researcher may find data to be highly sensitive, confidential or proprietary and therefore may be restricted in the subsequent utilization of certain materials.

CHAPTER II

LITERATURE REVIEW

Introduction

It is the primary purpose of the this chapter to introduce the reader to the comprehensive search of historical, technical and contemporary literature, including:

- A description of how literature was secured
- Validation of the research topic

How literature was secured

As part of the literature search it is imperative to describe the process that was utilized in order to illustrate to the reader the in-depth effort undertaken by the researcher. The researcher utilized the following sources:

- World-Wide Web, Building 6601, Room 144, Utah Avenue, Vandenberg AFB, CA 93437; and 1404 Swallow Court, Santa Maria, CA 93454
- Personal Library of Researcher, 1404 Swallow Court, Santa Maria, CA 93454

The actual information obtained from the above mentioned sources were,

- Internet, utilizing the keywords: National Missile Defense (59 hits), Ballistic Missile Defense Organization (32 hits), Theatre Missile Defense (35 hits) and Strategic Defense initiative (11 hits.)
- Personal Library, 1404 Swallow Court, Santa Maria, CA 93454

Validation of Research Topic

Major Krause, United States Air Force, details the birth/history of the anti-missile/strategic defense initiative defenses of this country and gives in-depth insight into early theater missile defense and anti-missile offensive actions, including the Bomarck,

Nike, Thor, Sprint/Spartan, Strategic Defense Initiative and the venerable Patriot missile from Desert Storm fame. In his thesis, From Theater Missile defense to Antimissile Offensive Actions, he details the birth of antimissile offensive actions from crude World War Two interdiction methods to current, and theorized antimissile options including the Airborne Laser, which is a high powered laser mounted on a Boeing 747 chassis. Additionally, Major Krause discusses a variety of programs under the various U.S. Military services, which make up the “catcher’s mitt theory”. This has been the underlying United States anti-missile defense theory for more than 40-years.

Theater missile and weapons of mass destruction proliferation creates a new set of challenges for the USAF in this multi-polar, post cold-war world. With the help of systems closing the time loop between the sensor and the shooter, the near term reinvigoration of USAF attack operations and associated Battle Management Command, Control and Communication Systems is possible. Once improved, attack options and Battle Management Command, Control and Communication Systems will effectively project national will and combat power while protecting our troops and allies with a strong offensive capability in a variety of situations.

(Krause, 1998, p.49)

Another study, which has had far reaching impact on the National Missile Defense program was funded by the Pentagon on the feasibility of the National Missile Defense system and to what extent the long-range system will work and how the research and development is progressing. Retired General Welch (United States Air Force) was commissioned by the Office of the Secretary of Defense to provide an unbiased overview of the National Missile Defense program and showcase its strengths and weaknesses.

His findings were highlighted the "Rushing to failure" theory that highlighted the fact that the Pentagon and the military-industrial complex is pushing forward the timetable on the eventual deployment of the National Missile Defense system without adequately testing the system. There has been a lack of ground testing of boosters, fully integrated flight tests with actual decoys. His findings were critical of the program management. However, there was no fundamental flaw in the technology or systems approach to solving the problem of intercepting long-range ICBMs with "hit to kill" weapons. The report highlighted flaws with procuring enough spares for full system tests, the lack of an established ground based tester, which would test the booster without engaging in flight and no conclusive way to conduct full system tests before operational deployment of the entire system.

IFT-3 and IFT-4 will rely heavily on Kwajalein Missile range (KMR) radars and equipment to support the EKV in flight. However, beginning with IFT-5, prototype versions of the Battle Management Command, Control and Communication Systems, X-band radar will be used making IFT-7 the first Integrated Systems test." (Welch Report, 2000, pg. 33). (As a side note IFT-7 was successful in "hitting and killing" a mock warhead launched from Vandenberg Air Force Base in California.)

The majority of the Welch Report is filled with recommendations on how to reduce the risk associated with the development program and how to effectively field a credible anti-missile option by 2007. However, as stated above the most compelling evidence has been the success of IFT-7 and IFT-8 using representative configuration of supporting hardware during the intercept tests.

The General Accounting Office published its "Cruise Missile Defense" paper and showed the fact that there is a need for a short-range missile defense from a cruise missile threat. This system has been designed to protect the American ground forces in foreign lands. Nowadays the need has been as acute than ever, with American troops fighting Taliban and Al Qaeda fighters in the rugged mountains of Afghanistan. Recent findings have shown that the Taliban and the Al Qaeda terrorist organization may have had the technology and/or the opportunity to make a "dirty nuclear bomb." Coupled with a crudely made cruise missile, the United States could be facing tremendous losses. This is why the need for a short-range missile defense is essential, and the technology will help the National Missile Shield. Also, this system is part of the layered systems approach to defending against missile attacks.

Current program timelines require the military services to field a fully capable cruise missile defense capability by 2010. And provide the capability to defend the nation against emerging threats in the near term. Each service is developing organic capability in order to address the cruise missile threat. This fragmented approach is due to the varying missions that each service is expected to fulfill. These unique systems will become part of multi-mission weapon systems that will provide defense against a wide range of threats. For example, the Navy Aegis cruiser is expected to engage theater ballistic missiles, aircraft, cruise missiles, surface targets and submarines while providing command and control functions over the entire navy carrier battle group.

Consequently, the U.S. Navy tested and scored a direct hit on a mock cruise missile during the first test of the Aegis missile defense systems. This demonstration was to prove that the advanced electronics package aboard the aegis cruisers could direct, track

and kill cruise missiles. This system will play into the catcher's mitt theory of providing long and short-range capabilities. (GAO, 1999)

Henry Hooper in his testimony to the House Subcommittee on National Security, Veterans Affairs and International Affairs Committee on Government Reform on 8 September 2000, provided the following synopsis of the Engineering Risk and the Anti-Ballistic Missile Treaty: "The U.S. Aerospace community has repeatedly met more daunting engineering challenges posed by the task of building effective ballistic missile defense. Consequently, there is little doubt, from an engineering perspective, that effective defenses can be built". (Hooper, 2000, p16)

Additionally, he concluded that better management practices would help ensure the on-time construction and deployment of the missile defense system. He cited Retired Air Force General Larry Welch's report to congress on the missile defense system and compared the missile defense system to other highly criticized development programs such as the manned space program. It was plagued by numerous very public test failures before a successful launch capability was achieved—just as had been the case in the unmanned space program before it. So it was also with the earlier development of the Air Force's Intercontinental Ballistic Missiles and the Navy's Submarine Launched Ballistic Missiles. (Hooper, 2000)

During April 2000 the Pentagon staffed an internal commission on the National Missile Defense test program. This report is more commonly referred to as The Coyle Report, prepared by the Special Investigations Division Committee on Government Reform, U.S. House of Representatives. The report covers the following areas: Origins of National Missile Defense; The current Administration's plan for National Missile

Defense; Costs of National Missile Defense; other administrative proposals to counter weapons of mass destruction; status of National Missile Defense; Flight-testing, Schedule and development issues and Global reactions to National Missile Defense. The area I want to concentrate on is the development of the National Missile Shield.

The Administration's National Missile Defense system is designed to shoot-down ICBMs as they travel through space. When an enemy missile is launched, the National Missile Defense system must detect it, accurately predict where it will be during the 30 or so minutes it will be in flight, determine which of the objects sailing through space toward the United States is the actual missile (as opposed to decoys designed to confuse sensors), and finally send a computer-guided interceptor to collide with the missile's warhead. To accomplish those tasks, National Missile Defense depends on a globe-spanning system of satellites, radars, communications systems, and battle management computers to launch and direct interceptors.

(Coyle Report, 2001, pg. 15)

The Bush Administrations plan calls for three distinct phases of development of the National Missile Shield. Each of the phases is referenced to as Capability 1, 2 and 3. These mark milestones in the program development and the evolution of the entire National Missile Defense program. This is outlined in figure 2 below, which shows

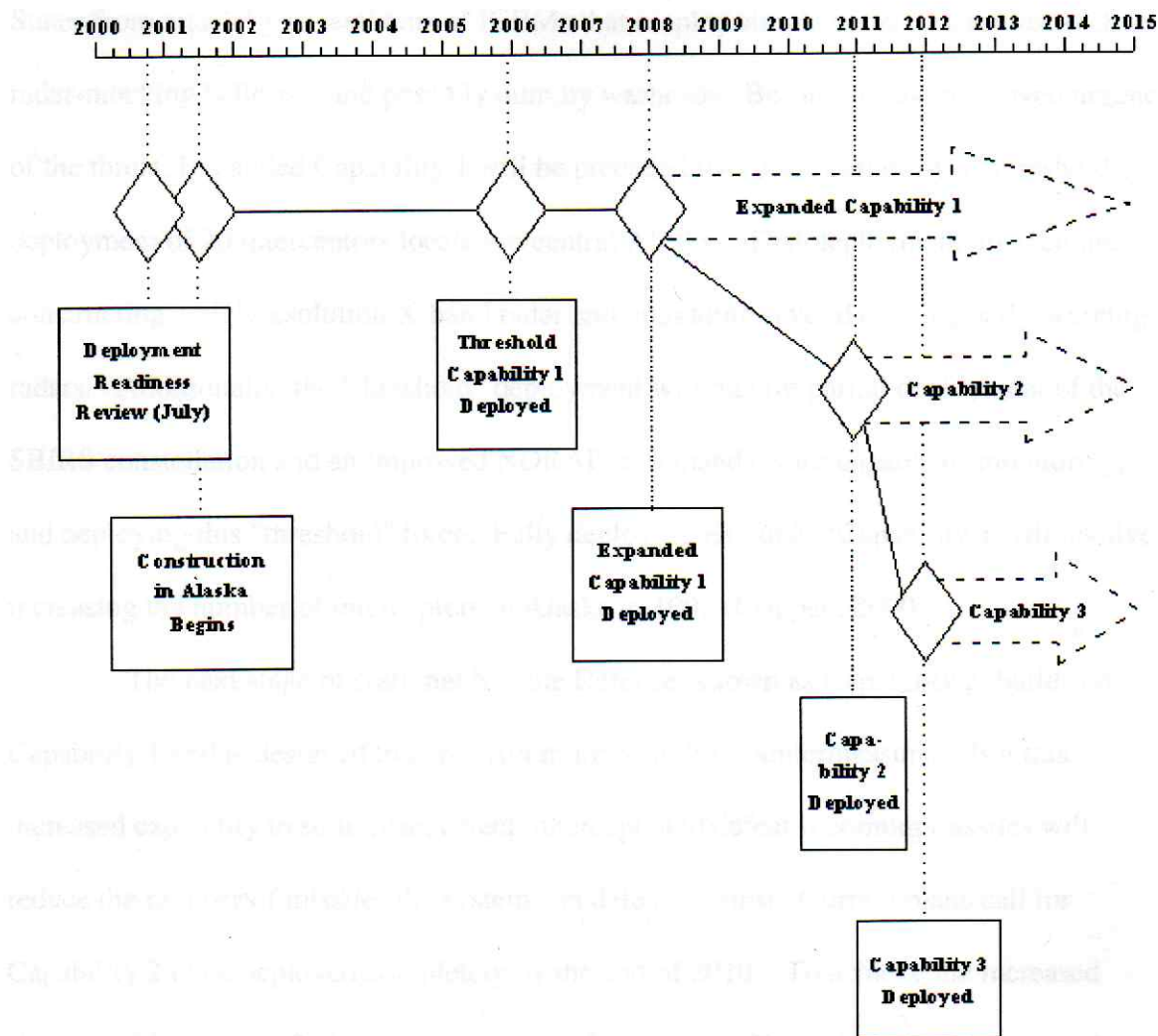


FIGURE 2. National Missile Defense Milestone Chart. Note. From The Coyle Report (p.8) by Dan Crippen, 2000, Washington D.C.

deployment, construction in Alaska, Capability 1 deployed in 2006 and finally Capability 2 being deployed in 2012. (Crippen, 2000)

The Bush Administration's plan for developing National Missile Defense calls for the first stage, Expanded Capability 1. To be fully deployed by the end of fiscal year 2007 (Figure 2). After Capability 1 is deployed, it is intended to defend the entire United

States from attack by several tens of ICBMs that employ simple countermeasures, such as radar-mocking balloons, and possibly dummy warheads. Because of the perceived urgency of the threat, Expanded Capability 1 will be preceded two years earlier by a "threshold" deployment of 20 interceptors located in central Alaska. That deployment also requires constructing a high resolution X-band radar and upgrading several existing early-warning radars. Additionally, the "threshold" deployment will require partial deployment of the SBIRS constellation and an improved NORAD command center capable of monitoring, and deploying this "threshold" force. Fully deploying Expanded Capability 1 will involve increasing the number of interceptors in Alaska to 100. (Crippen, 2000)

The next stage of National Missile Defense, known as Capability 2, builds on Capability 1 and is designed to cope with more complex countermeasures. But this increased capability to scrutinize, detect, intercept, and defeat-incoming missiles will reduce the number of missiles the system can defend against. Current plans call for Capability 2 to be deployed completely by the end of 2010. To achieve the increased abilities of Capability 2, the system would add three more X-band radars at various sites around the world and more facilities to communicate with interceptors in flight...By the time it was deployed, Capability 2 would have the full benefit of the entire SBIRS-high and SBIRS-low satellite constellation. According to the Bush Administration's plan under Capability 2, SBIRS-high would continue to supply early warning information to the National Missile Defense system as well as to the rest of the U.S. strategic forces. Although SBIRS-low is intended to continuously buttress the National Missile Defense system, it will also support theater missile defense (systems designed to defend areas outside of the United States from relatively short-range missiles). Both the precise

tracking of SBIRS-low and its ability to distinguish warheads and decoys should significantly aid theater missile defenses. Unlike National Missile Defense, theater missile defense systems are limited in both the area they protect and the length of time for which they are to be deployed. (Crippen, 2000)

The final level of National Missile Defense deployment is Capability 3, which includes all the assets of Capability 2 plus 150 additional interceptors, more radars, another communications facility, and improved software for each systems' components. This stage would combine the capabilities of the two earlier stages by defending the country from several tens of incoming missiles with complex countermeasures. Some of the additional interceptors would be stationed at a second site, currently planned for Grand Forks, North Dakota. That would improve the system's coverage of the United States by placing interceptors closer to the East Coast. From there, they could attack warheads originating in the Middle East at farther distances from the United States—and thus earlier in the warhead's flight—than interceptors based in Alaska could. (Crippen, 2000, pg. 55).

The Coalition to Reduce Nuclear Dangers/Council for a Livable World Education Fund published, Pushing the Limits—Decision on National Missile Defense. In this Article they address various subjects, one of them the security of the United States in the modern -era and how the National Missile Defense system will play into this countries security.

For forty years, the United States has lived with the threat of attack by missiles armed with chemical, biological or nuclear weapons. Russia's arsenal of thousands of nuclear warheads on long-range missiles could still destroy the United States in

under an hour. However, missile defense advocates do not claim the defense they propose could deal with a large Russian attack. They cite instead new threats to justify a first phase of missile defense; from North Korea, which recently froze its meager ballistic missile testing program; Iran, although experts are divided on whether its nascent missile efforts will be able to threaten the United States within 10 years. Iraq, which is currently under international sanctions that sharply hinder its ability to develop new missiles. If the United States deploys a missile defense in an attempt to counter a handful of missiles North Korea might build, it could exacerbate tensions with heavily armed Russia and instigate a vigorous Chinese build-up. Even further, it is unclear why these countries would commit an almost certainly suicidal attack on the United States. (Young, 2000, pg 2-2)

The National Missile Defense Independent Review Team (IRT) conducted a Executive summary of the program at the request of the Secretary of Defense during 2000. The executive summary had some key findings, which I would like to discuss now and show their relevance towards proving my argument that National Missile Defense can protect the United States against a rogue missile attack.

The IRT concluded that the technical capability to develop and field the limited system to meet the defined Capability 1 threat is currently available. There has been important program progress in the past year, which includes a successful intercept, demonstrated integration of several prototype system elements, and continued development of simulations and laboratories for ground testing. Regarding the first decision milestone, the Deployment Readiness Review, the criteria is as follows. Two successful intercepts, one of which must be an integrated system test. (A test that

includes integrating the multiple elements of the system—radar, ground based interceptor, command and control, communication). IFT-4 was to be the first possible integrated systems test. It did demonstrate integration of the system to a degree not previously achieved. (IRT, 2000) However, as another side note, IFT5-8 was successful in hitting the target warhead and did prove systems integration.

The Defense Press Service provides monthly updates on various programs to include the National Missile Defense testing and integration of the systems. During December 2001, the Defense Pres Service reported the following:

The Ballistic Missile Defense Organization scored another bulls-eye Dec. 3 in a test of the “kill vehicle,” officials announced. This was the third successful intercept in five tests of the system. This test—IFT-7 was identical to a previous test in July 2001. In the test BMDO launched a modified Minuteman ICBM as a target from Vandenberg AFB, CA, at 9:59 p.m. EST. The launch was supposed to go at 9 p.m., but it was delayed by weather. About 20 minutes after the launch from California, officials launched an interceptor from the Ronald Reagan Missile Range at the Kwajalein Atoll in the Pacific. The kill vehicle separated from its rocket booster more than 1,400 miles from the target warhead. After separation, the vehicle used its on-board infrared and visual sensors to locate and track the target. The sensors were augmented by X-Band radar data from the Battle Management Command, Control and Communications facilities in Colorado Springs, Colo., and Kwajalein Atoll. The sensors were able to distinguish between the target, the shroud enclosing the missile, and a decoy deployed in the test. The interceptor hit the target more than 140 miles above the earth during the midcourse phase of the warhead’s flight. The

closing speed for the hit to kill intercept was "in excess of 15,000 miles per hour.

(Garamone, 2001, pg.1)

After the conclusion of the Gulf War, the Israeli Defense Forces set about on a robust research, development and testing program for an organic missile defense system. They applied the same tenacity to this project as they do with every military under taking. And the results have become apparent, Israel has a ballistic missile defense system. Code named Arrow, this system is on-duty and ready for any new ballistic threats whether from Iraq or from Syria. Henry F. Cooper, compares the Israeli Arrow anti missile program to the up-start National Missile Defense program that has still flight and integration testing to go before fielding.

For less than \$2 billion (three-quarters of which came from U.S. taxpayers), Israel now has its own homeland defense against ballistic missiles—a capability only shared with Russia. With the right management and technical skills, the United States Navy can provide our earliest opportunity to join Israel and Russia with our own homeland ballistic missile defense system and begin protecting United States cities against ballistic missiles. The Navy should adopt an engineering-management approach like the one they used during the Polaris project, which produced our first strategic nuclear submarine and its associated submarine-launched ballistic missiles in only four years in the later '50s and early '60s. (Hooper, 2001, pg. 17)

The Pacific Life Research Center (PLRC) has published numerous articles on the National Missile Defense system. These articles discuss the fundamental system components and how the system is supposed to work and what the system background. The three articles are: Ballistic Missile Defense 1: BMD Structure, Battle Management,

and Sensors, Ballistic Missile Defense 2: Boost Phase Interceptors, and From Star Wars to Scud Buster: A Background Paper on Ballistic Missile Defense.

The concept of defending against missiles was conceived in the 1940s due to the German V-1 and V-2 attacks on London. During the 1960s this defensive system became more publicly known and refined after ballistic missiles were introduced as delivery methods for nuclear warheads. It was at first called ballistic missile defense (BMD) and the interceptor used to counter ballistic missiles was called an anti ballistic missile (Aldridge, 2001)

Decades later and tens of billions dollars poorer, the United States is still attempting to develop a workable ABM system. In 1983 President Ronald Reagan initiated the Strategic Defense Initiative (SDI, or Star Wars) with an increase of the defense budget. From 1985 through 2001 there has been \$58 billion spent on different defensive measures. Under Reagan Star Wars program there was envisioned a layered defense where various weapons would intercept and destroy hostile missiles in the boost-, midcourse-, and terminal phases. The three intercept layers of defense, which are being studied are 1) Boost-Phase Interceptors would destroy hostile missiles during their ascent out of the earth's stratosphere. 2) Midcourse Phase interceptors would target and destroy the missiles during their final stage and while the warheads are coasting through space. 3) Terminal-Phase interceptors would target and destroy the warheads as they are reentering the earth's atmosphere and bearing down on their targets. (Aldridge, 2001)

Realizing that the compressed deployment schedule so avidly championed during his election campaign did not allow time to solve the overwhelming objections of US allies. And to negotiate the differences with Russia on the constrictive ABM Treaty,

President Bush changed direction to an aggressive research, development testing, and evaluation program. This has opened a gamut of technologies to investigate, test and field. Along with the name change, the spectrum of BMD weapons now being addressed closely emulates the Star Wars of the 1980s. It introduces open-ended spending that will cause BMD spending to soar, possibly over \$100 billion. (Aldridge, 2001, pg. 2)

The boost-phase is the time the first phase of a missiles flight, when the booster rocket motors are burning and accelerating the missile to gain altitude and speed. It lasts from 3 minutes to 5 minutes, depending on the size of strategic missile. For short-range missiles it can be as low as 1-2 minutes. To intercept a missile during this phase, it must be done within 1-5 minutes. Intercepting in the boost phase does have its advantages even though the target is moving fast and presents a hard target to kill due to its speed and shortness of opportunity. First of all, it would destroy the missile before it has had time to dispense multiple warheads or chemical-biological canisters. The debris would likely fall back onto the hostile country's own territory. Finally, a boost phase intercept would be effective against missiles of any range. The Bush Administration's boost phase intercept program has three segments—Airborne Laser, Sea-based boost, and Space Laser Experiment. Within a minute the early warning radars and space-based sensors detect and characterize the hostile missile's launch and proposed path and possibly its target. The ABL, which would be patrolling above the cloud at about 40,000 feet, has to then locate, point, and fire its high powered laser at the missile. It would train the high-energy laser beam on the missiles pressurized rocket motors or fuel tanks. After a few seconds the missile bursts from the intense heat. (Aldridge, 2001)

Midcourse starts after the rocket motors burn out and before the warheads start

reentering the earth's atmosphere. It is the time when the warhead separates from the missile structure or, in the case of multiple warheads and chemical-biological weapons, when the warheads or canisters separate from the deployment platform. This is also the time when penetration aids such as chaff and decoys would be deployed in order to confuse radar systems. The midcourse phase of a ballistic missile's flight lasts about 20-minutes—less time for shorter missiles. It is the longest time of any of the three phases of a missiles' flight. The systems being pursued to shoot down missiles during this phase are the Ground-based Midcourse Segment and the Sea-based Midcourse Segment. (Aldridge, 2001)

NOTE: Since the beginning of this research project, there has been successful progress on the various elements of the program to include two more successful intercepts, two successful radar interrogations of boosters, categorization of test warheads reentering the earth's atmosphere and one successful prototype booster launch. However, there has been set backs as well. A prototype booster disintegrated upon launch from its silo and was destroyed 400 yards off shore.

It has been the intention of the researcher to present to the reader with the history of the ballistic missile defense program, its noteworthy accomplishments and failures. Additionally, the researcher attempted to highlight the process and the results of a comprehensive literature search including methods by which the literature was obtained. And how this validated the research topic. The researcher believes him to be embarking into a new area of original research.

The study will now proceed on to Chapter III where the researcher will discuss in detail the analysis of data and how it supported his argument.

CHAPTER III

RESULTS, FINDINGS & CONCLUSIONS

In Chapter Three the researcher will report on the results of findings obtained through exhaustive research followed by an effort to correlate all data received from the literature search, described in Chapter Two. Conclusions will be developed by the researcher at the end of this chapter in an effort to validate the project's hypothesis, which is,

“Can the National Missile Shield can protect the Continental United States against a rogue missile threat/attack.”

This chapter will begin with the literature search; the researcher can offer the following conclusions:

- The current technology is sufficiently advanced to hit and kill incoming ballistic missile warheads.
- Current tests do have a high percentage of system integration, thus reflecting the capability of the system to work together and protect the United States.
- The Ballistic Missile Defense Organization's “Catcher's Mitt” theory on missile defense can work.
- The ballistic missile threat from North Korea, Iran, Iraq and China is credible enough to warrant the pursuit of this missile defense.

It would appear, based on the conclusions arrived at, that the project's research question which is, “Can the National Missile Shield can protect the Continental United States against a rogue missile threat/attack” has been validated, but the project's null hypothesis, which is, “The National Missile Shield cannot protect the Continental United States against a rogue missile threat/attack” has not been validated.

The researcher will now conclude this project in Chapter Four by identifying a series of recommendations for future review, consideration and possible implementation.

CHAPTER IV

RECOMMENDATIONS

This research project will conclude with a series of recommendations preceded by some limitations that have been experienced during the conduct of this study. The following the recommendations, the researcher will attempt to identify some implications that could occur as a result of this study.

Limitations

Prior to making a series of recommendations, the following limitations directly or indirectly may have had an impact on the overall results. Limitations include:

- The researcher was unable to dedicate an uninterrupted period of time to this project.

Because of regular employment circumstances such as a job of increased responsibilities and an ongoing inspection from higher headquarters inspector general along with a compresses launch schedule, which includes five-test launches of intercontinental ballistic missiles, the researcher could not complete the study in a timely manner.

Further developments in the national security posture of the United States of America have had an impact on this study.

Recommendations

Based on the results, findings and conclusions identified Chapter IV, the researcher is offering the following recommendations for review, consideration and possible implementation:

The researcher recommends further testing of the research question every six months to determine if the system has improvement areas and see how the system performs under an increased operations tempo.

- The researcher recommends further studies into the effects of counter measures in testing and the development of testing areas outside of the Pacific Rim area. This will help with the question of defending the East Coast.
- The researcher recommends the build-up of spares, to facilitate more testing and ground based testing of the boosters and the EKV. Additionally, the researcher recommends computer modeling to simulate test launches in adverse weather.

If all or some of these recommendations are actually instituted, the researcher could foresee such implications as following:

- The flight-testing and end-to-end testing of the entire National Missile Defense system will raise the public confidence in the system's ability to defend the United States against a rogue missile threat.
- Implementation/fielding of a new theater missile defense system, which can be proven before deployment but tested under battlefield conditions.
- The "rogue nations" attempting to hit the United States with a ballistic missile will increase with the completion of the National Missile Defense system.
- Advanced counter measures on ballistic missiles will continue the research and development of the National Missile Defense system and will make the cost increasingly expensive.

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